

CH 3 — Continuity

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1.1 Continuous Functions

Definition — Continuity

We say that $f(x, y)$ is continuous at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

If f is continuous at every point in $D \subseteq \mathbb{R}^2$, then f is continuous on D . It requires:

1. $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists
2. $f(a, b)$ is defined
3. $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$

Definition — Continuity Theorems

If $f(x, y)$ and $g(x, y)$ are scalar functions and $(x, y) \in D(f) \cap D(g)$, then:

1. $f + g$ and $f \cdot g$ are continuous at (a, b)
2. $\frac{f}{g}$ is also continuous at (a, b)

Composite Functions

For scalar functions $g(t)$ and $f(x, y)$, the composite function is $g \circ f(x, y) = g(f(x, y))$ for all $(x, y) \in \text{dom } f$ s.t. $f(x, y) \in \text{dom } g$

3. If $f(x, y)$ is continuous at (a, b) and $g(t)$ is continuous at $f(a, b)$, then $g \circ f$ is continuous at (a, b)

Example:

1. Is $f(x, y) = y \sin\left(\frac{x}{y}\right)$ continuous?

Notice that $\frac{x}{y}$ is continuous in $\mathbb{R} \setminus \{0\} \forall (x, y) \in \mathbb{R}^2$ s.t. $y \neq 0$

We know $\sin t$ is continuous $\forall t \in \mathbb{R}$, so $\sin\left(\frac{x}{y}\right)$ is continuous $\forall (x, y) \in \mathbb{R}^2$ s.t. $y \neq 0$

By continuity theorems, $y \sin\left(\frac{x}{y}\right)$ is continuous wherever both functions are continuous.

Thus $f(x, y)$ is continuous on $\{(x, y) \in \mathbb{R}^2 : y \neq 0\}$

2. Evaluate $\lim_{(x,y) \rightarrow (1,0)} \frac{x^2 \ln x - x^3 \cos y}{x^3 + y^5}$

By continuity theorems, this is continuous at $(1, 0)$, therefore $\lim_{(x,y) \rightarrow (1,0)} \frac{x^2 \ln x - x^3 \cos y}{x^3 + y^5} = -1$

 **Tip – Applications with Squeeze Theorem** When using the Squeeze Theorem to prove a limit exists, we should try to find an upper bound $B(x, y)$ which is continuous so that

$$\lim_{(x,y) \rightarrow (a,b)} B(x, y)$$

is easy to evaluate

When the continuity theorems don't apply, then it is needed to use the definition of continuity.

Example:

$$1. f(x, y) = \begin{cases} \frac{x}{\sqrt{x^2+y^2}} & (x,y) \neq 0 \\ 1 & (x,y) = (0,0) \end{cases}$$

By continuity theorems, $f(x, y)$ is continuous $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$

Check if $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1$

Notice that if $x = 0 \implies \lim_{(x,y) \rightarrow (0,0)} f(0, y) = \lim_{y \rightarrow 0} f(y) = \lim_{y \rightarrow 0} \frac{0}{\sqrt{y^2}} = 0 \neq 1$

If the limit exists, it must be 0, so it is not continuous at $(0, 0)$

$$2. f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x,y) \neq 0 \\ 0 & (x,y) = (0,0) \end{cases}$$

By continuity theorems, $f(x, y)$ is continuous $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$

Check if $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$

Try $y = mx : \lim_{(x,y) \rightarrow (0,0)} f(x, mx) = \lim_{x \rightarrow 0} \frac{2m}{1+m^2} = \frac{2m}{1+m^2}$

The limit depends on m , so the limit does not exist. Therefore, f is not continuous at $(0, 0)$